Equation Sample 6

Dataset-expansion sample

1 Derivation

The following display is drawn from a source-backed image-to-LaTeX benchmark and reproduced verbatim.

$$\begin{aligned}
&G_{c_p}(b,c_p)\\
&\qquad= -W^{-1}\Big(\varphi(b)\int_0^b \psi(z,0)\frac{\partial}{\partial c_p}\pi_1(z,c_p)m'(z)dz + \psi(b,0)\int_b^\infty \varphi(z)\frac{\partial}{\partial c_p}\pi_1(z,c_p)m'(z)dz\Big),\\
&G_{bc_p}(b,c_p)\\
&\qquad= -W^{-1}\Big(\varphi'(b)\int_0^b \psi(z,0)\frac{\partial}{\partial c_p}\pi_1(z,c_p)m'(z)dz + \varphi(b)\psi(b,0)\frac{\partial}{\partial c_p}\pi_1(b,c_p)m'(b)\\
&\qquad\qquad + \psi'(b,0)\int_b^\infty \varphi(z)\frac{\partial}{\partial c_p}\pi_1(z,c_p)m'(z)dz - \varphi(b)\psi(b,0)\frac{\partial}{\partial c_p}\pi_1(b,c_p)m'(b)\Big)\\
&\qquad= -W^{-1}\Big(\varphi'(b)\int_0^b \psi(z,0)\frac{\partial}{\partial c_p}\pi_1(z,c_p)m'(z)dz + \psi'(b,0)\int_b^\infty \varphi(z)\frac{\partial}{\partial c_p}\pi_1(z,c_p)m'(z)dz\Big).
\end{aligned}$$